

Code for QSS Chapter 2: Causality

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First Printing

Section 2.1: Racial Discrimination in the Labor Market

```
resume <- read.csv("resume.csv")

dim(resume)

## [1] 4870    4

head(resume)

##   firstname    sex  race call
## 1  Allison female white    0
## 2  Kristen female white    0
## 3  Lakisha female black    0
## 4  Latonya female black    0
## 5   Carrie female white    0
## 6     Jay   male white    0

summary(resume)

##   firstname      sex      race      call
## Tamika : 256   female:3746   black:2435   Min.    :0.00000
## Anne   : 242   male  :1124   white:2435   1st Qu.:0.00000
## Allison: 232                                     Median :0.00000
## Latonya: 230                                     Mean   :0.08049
## Emily  : 227                                     3rd Qu.:0.00000
## Latoya : 226                                     Max.   :1.00000
## (Other):3457

race.call.tab <- table(race = resume$race, call = resume$call)
race.call.tab

##           call
## race         0     1
## black 2278 157
## white 2200 235

addmargins(race.call.tab)

##           call
## race         0     1  Sum
## black 2278 157 2435
## white 2200 235 2435
## Sum   4478 392 4870

## overall callback rate: total callbacks divided by the sample size
sum(race.call.tab[, 2]) / nrow(resume)

## [1] 0.08049281
```

```
## callback rates for each race
race.call.tab[1, 2] / sum(race.call.tab[1, ]) # black

## [1] 0.06447639

race.call.tab[2, 2] / sum(race.call.tab[2, ]) # white

## [1] 0.09650924

race.call.tab[1, ] # the first row

##      0      1
## 2278  157

race.call.tab[, 2] # the second column

## black white
##    157    235
mean(resume$call)

## [1] 0.08049281
```

Section 2.2: Subsetting the Data in R

```
class(TRUE)

## [1] "logical"

as.integer(TRUE)

## [1] 1

as.integer(FALSE)

## [1] 0

x <- c(TRUE, FALSE, TRUE) # a vector with logical values

mean(x) # proportion of TRUEs

## [1] 0.6666667

sum(x) # number of TRUEs

## [1] 2

FALSE & TRUE

## [1] FALSE

TRUE & TRUE

## [1] TRUE

TRUE | FALSE

## [1] TRUE

FALSE | FALSE

## [1] FALSE
```

```
TRUE & FALSE & TRUE

## [1] FALSE

(TRUE | FALSE) & FALSE # the parentheses evaluate to TRUE

## [1] FALSE

TRUE | (FALSE & FALSE) # the parentheses evaluate to FALSE

## [1] TRUE

TF1 <- c(TRUE, FALSE, FALSE)
TF2 <- c(TRUE, FALSE, TRUE)
TF1 | TF2

## [1] TRUE FALSE TRUE

TF1 & TF2

## [1] TRUE FALSE FALSE
```

Section 2.2.2: Relational Operators

```
4 > 3

## [1] TRUE

"Hello" == "hello" # R is case-sensitive

## [1] FALSE

"Hello" != "hello"

## [1] TRUE

x <- c(3, 2, 1, -2, -1)
x >= 2

## [1] TRUE TRUE FALSE FALSE FALSE

x != 1

## [1] TRUE TRUE FALSE TRUE TRUE

## logical conjunction of two vectors with logical values
(x > 0) & (x <= 2)

## [1] FALSE TRUE TRUE FALSE FALSE

## logical disjunction of two vectors with logical values
(x > 2) | (x <= -1)

## [1] TRUE FALSE FALSE TRUE TRUE

x.int <- (x > 0) & (x <= 2) # logical vector
x.int

## [1] FALSE TRUE TRUE FALSE FALSE

mean(x.int) # proportion of TRUEs

## [1] 0.4
```

```
sum(x.int) # number of TRUEs
```

```
## [1] 2
```

Section 2.2.3: Subsetting

```
## callback rate for black-sounding names  
mean(resume$call[resume$race == "black"])
```

```
## [1] 0.06447639
```

```
## race of first 5 observations  
resume$race[1:5]
```

```
## [1] white white black black white  
## Levels: black white
```

```
## comparison of first 5 observations  
(resume$race == "black")[1:5]
```

```
## [1] FALSE FALSE TRUE TRUE FALSE
```

```
dim(resume) # dimension of original data frame
```

```
## [1] 4870 4
```

```
## subset blacks only  
resumeB <- resume[resume$race == "black", ]  
dim(resumeB) # this data.frame has fewer rows than the original data.frame
```

```
## [1] 2435 4
```

```
mean(resumeB$call) # callback rate for blacks
```

```
## [1] 0.06447639
```

```
## keep "call" and "firstname" variables  
## also keep observations with black female-sounding names  
resumeBf <- subset(resume, select = c("call", "firstname"),  
                   subset = (race == "black" & sex == "female"))  
head(resumeBf)
```

```
##   call firstname  
## 3    0   Lakisha  
## 4    0   Latonya  
## 8    0    Kenya  
## 9    0   Latonya  
## 11   0    Aisha  
## 13   0    Aisha
```

```
## ## an alternative syntax with the same results  
## resumeBf <- resume[resume$race == "black" & resume$sex == "female",  
##                    c("call", "firstname")]  
## black male  
resumeBm <- subset(resume, subset = (race == "black" & (sex == "male"))  
## white female  
resumeWf <- subset(resume, subset = (race == "white" & (sex == "female"))  
## white male
```

```

resumeWm <- subset(resume, subset = (race == "white") & (sex == "male"))
## racial gaps
mean(resumeWf$call) - mean(resumeBf$call) # among females

## [1] 0.03264689

mean(resumeWm$call) - mean(resumeBm$call) # among males

## [1] 0.03040786

```

Section 2.2.4: Simple Conditional Statements

```

resume$BlackFemale <- ifelse(resume$race == "black" &
                             resume$sex == "female", 1, 0)
table(race = resume$race, sex = resume$sex,
      BlackFemale = resume$BlackFemale)

## , , BlackFemale = 0
##
##      sex
## race  female male
##  black      0  549
##  white  1860  575
##
## , , BlackFemale = 1
##
##      sex
## race  female male
##  black  1886    0
##  white    0    0

```

Section 2.2.5: Factor Variables

```

resume$type <- NA
resume$type[resume$race == "black" & resume$sex == "female"] <- "BlackFemale"
resume$type[resume$race == "black" & resume$sex == "male"] <- "BlackMale"
resume$type[resume$race == "white" & resume$sex == "female"] <- "WhiteFemale"
resume$type[resume$race == "white" & resume$sex == "male"] <- "WhiteMale"

## check object class
class(resume$type)

## [1] "character"

## coerce new character variable into a factor variable
resume$type <- as.factor(resume$type)
## list all levels of a factor variable
levels(resume$type)

## [1] "BlackFemale" "BlackMale"   "WhiteFemale" "WhiteMale"

## obtain the number of observations for each level
table(resume$type)

```

```
##
## BlackFemale BlackMale WhiteFemale WhiteMale
##      1886      549      1860      575
tapply(resume$call, resume$type, mean)

## BlackFemale BlackMale WhiteFemale WhiteMale
## 0.06627784 0.05828780 0.09892473 0.08869565

## turn first name into a factor variable
resume$firstname <- as.factor(resume$firstname)
## compute callback rate for each first name
callback.name <- tapply(resume$call, resume$firstname, mean)
## sort the result in the increasing order
sort(callback.name)
```

```
##      Aisha      Rasheed      Keisha      Tremayne      Kareem      Darnell
## 0.02222222 0.02985075 0.03825137 0.04347826 0.04687500 0.04761905
##      Tyrone      Hakim      Tamika      Lakisha      Tanisha      Todd
## 0.05333333 0.05454545 0.05468750 0.05500000 0.05797101 0.05882353
##      Jamal      Neil      Brett      Geoffrey      Brendan      Greg
## 0.06557377 0.06578947 0.06779661 0.06779661 0.07692308 0.07843137
##      Emily      Anne      Jill      Latoya      Kenya      Matthew
## 0.07929515 0.08264463 0.08374384 0.08407080 0.08673469 0.08955224
##      Latonya      Leroy      Allison      Ebony      Jermaine      Laurie
## 0.09130435 0.09375000 0.09482759 0.09615385 0.09615385 0.09743590
##      Sarah      Meredith      Carrie      Kristen      Jay      Brad
## 0.09844560 0.10160428 0.13095238 0.13145540 0.13432836 0.15873016
```

Section 2.3: Causal Effects and the Counterfactual

```
resume[1, ]

##  firstname sex race call BlackFemale type
## 1  Allison female white 0 0 WhiteFemale
```

Section 2.4: Randomized Controlled Trials

Section 2.4.1: The Role of Randomization

Section 2.4.2: Social Pressure and Voter Turnout

```
social <- read.csv("social.csv") # load the data

summary(social) # summarize the data

##      sex      yearofbirth      primary2004      messages
## female:152702 Min. :1900 Min. :0.0000 Civic Duty: 38218
## male :153164 1st Qu.:1947 1st Qu.:0.0000 Control :191243
##      Median :1956 Median :0.0000 Hawthorne : 38204
##      Mean :1956 Mean :0.4014 Neighbors : 38201
##      3rd Qu.:1965 3rd Qu.:1.0000
```

```
##           Max.      :1986    Max.      :1.0000
##   primary2006      hhsizes
##   Min.      :0.0000    Min.      :1.000
##   1st Qu.:0.0000    1st Qu.:2.000
##   Median :0.0000    Median :2.000
##   Mean      :0.3122    Mean      :2.184
##   3rd Qu.:1.0000    3rd Qu.:2.000
##   Max.      :1.0000    Max.      :8.000

## turnout for each group
tapply(social$primary2006, social$messages, mean)

## Civic Duty      Control Hawthorne Neighbors
## 0.3145377 0.2966383 0.3223746 0.3779482

## turnout for control group
mean(social$primary2006[social$messages == "Control"])

## [1] 0.2966383

## subtract control group turnout from each group
tapply(social$primary2006, social$messages, mean) -
  mean(social$primary2006[social$messages == "Control"])

## Civic Duty      Control Hawthorne Neighbors
## 0.01789934 0.00000000 0.02573631 0.08130991

social$age <- 2006 - social$yearofbirth # create age variable
tapply(social$age, social$messages, mean)

## Civic Duty      Control Hawthorne Neighbors
## 49.65904 49.81355 49.70480 49.85294

tapply(social$primary2004, social$messages, mean)

## Civic Duty      Control Hawthorne Neighbors
## 0.3994453 0.4003388 0.4032300 0.4066647

tapply(social$hhsizes, social$messages, mean)

## Civic Duty      Control Hawthorne Neighbors
## 2.189126 2.183667 2.180138 2.187770
```

Section 2.5: Observational Studies

Section 2.5.1: Minimum Wage and Unemployment

```
minwage <- read.csv("minwage.csv") # load the data

dim(minwage) # dimension of data

## [1] 358 8

summary(minwage) # summary of data

##           chain           location    wageBefore    wageAfter
## burgerking:149    centralNJ: 45    Min.      :4.250    Min.      :4.250
```

```
## kfc      : 75  northNJ :146  1st Qu.:4.250  1st Qu.:5.050
## roys     : 88  PA      : 67  Median :4.500  Median :5.050
## wendys   : 46  shoreNJ : 33  Mean   :4.618  Mean   :4.994
##          :    southNJ : 67  3rd Qu.:4.987  3rd Qu.:5.050
##          :          :    Max.   :5.750  Max.   :6.250
## fullBefore fullAfter partBefore partAfter
## Min.   : 0.000  Min.   : 0.000  Min.   : 0.00  Min.   : 0.00
## 1st Qu.: 2.125  1st Qu.: 2.000  1st Qu.:11.00  1st Qu.:11.00
## Median : 6.000  Median : 6.000  Median :16.25  Median :17.00
## Mean   : 8.475  Mean   : 8.362  Mean   :18.75  Mean   :18.69
## 3rd Qu.:12.000  3rd Qu.:12.000  3rd Qu.:25.00  3rd Qu.:25.00
## Max.   :60.000  Max.   :40.000  Max.   :60.00  Max.   :60.00

## subsetting the data into two states
minwageNJ <- subset(minwage, subset = (location != "PA"))
minwagePA <- subset(minwage, subset = (location == "PA"))

## proportion of restaurants whose wage is less than $5.05
mean(minwageNJ$wageBefore < 5.05) # NJ before

## [1] 0.9106529
mean(minwageNJ$wageAfter < 5.05) # NJ after

## [1] 0.003436426
mean(minwagePA$wageBefore < 5.05) # PA before

## [1] 0.9402985
mean(minwagePA$wageAfter < 5.05) # PA after

## [1] 0.9552239
## create a variable for proportion of full-time employees in NJ and PA
minwageNJ$fullPropAfter <- minwageNJ$fullAfter /
  (minwageNJ$fullAfter + minwageNJ$partAfter)
minwagePA$fullPropAfter <- minwagePA$fullAfter /
  (minwagePA$fullAfter + minwagePA$partAfter)

## compute the difference in means
mean(minwageNJ$fullPropAfter) - mean(minwagePA$fullPropAfter)

## [1] 0.04811886
```

Section 2.5.2: Confounding Bias

```
prop.table(table(minwageNJ$chain))

##
## burgerking      kfc      roys      wendys
## 0.4054983 0.2233677 0.2508591 0.1202749

prop.table(table(minwagePA$chain))

##
## burgerking      kfc      roys      wendys
```

```
## 0.4626866 0.1492537 0.2238806 0.1641791
## subset Burger King only
minwageNJ.bk <- subset(minwageNJ, subset = (chain == "burgerking"))
minwagePA.bk <- subset(minwagePA, subset = (chain == "burgerking"))

## comparison of full-time employment rates
mean(minwageNJ.bk$fullPropAfter) - mean(minwagePA.bk$fullPropAfter)

## [1] 0.03643934
minwageNJ.bk.subset <-
  subset(minwageNJ.bk, subset = ((location != "shoreNJ") &
                                (location != "centralNJ")))

mean(minwageNJ.bk.subset$fullPropAfter) - mean(minwagePA.bk$fullPropAfter)

## [1] 0.03149853
```

Section 2.5.3: Before-and-After and Difference-in-Differences Designs

```
## full-time employment proportion in the previous period for NJ
minwageNJ$fullPropBefore <- minwageNJ$fullBefore /
  (minwageNJ$fullBefore + minwageNJ$partBefore)

## mean difference between before and after the minimum wage increase
NJdiff <- mean(minwageNJ$fullPropAfter) - mean(minwageNJ$fullPropBefore)
NJdiff

## [1] 0.02387474
## full-time employment proportion in the previous period for PA
minwagePA$fullPropBefore <- minwagePA$fullBefore /
  (minwagePA$fullBefore + minwagePA$partBefore)
## mean difference between before and after for PA
PAdiff <- mean(minwagePA$fullPropAfter) - mean(minwagePA$fullPropBefore)
## difference-in-differences
NJdiff - PAdiff

## [1] 0.06155831
## full-time employment proportion in the previous period for PA
minwagePA$fullPropBefore <- minwagePA$fullBefore /
  (minwagePA$fullBefore + minwagePA$partBefore)
## mean difference between before and after for PA
PAdiff <- mean(minwagePA$fullPropAfter) - mean(minwagePA$fullPropBefore)
## difference-in-differences
NJdiff - PAdiff

## [1] 0.06155831
```

Section 2.6: Descriptive Statistics for a Single Variable

Section 2.6.1: Quantiles

```
## cross-section comparison between NJ and PA
median(minwageNJ$fullPropAfter) - median(minwagePA$fullPropAfter)

## [1] 0.07291667

## before and after comparison
NJdiff.med <- median(minwageNJ$fullPropAfter) -
  median(minwageNJ$fullPropBefore)
NJdiff.med

## [1] 0.025

## median difference-in-differences
PAdiff.med <- median(minwagePA$fullPropAfter) -
  median(minwagePA$fullPropBefore)
NJdiff.med - PAdiff.med

## [1] 0.03701923

## summary shows quartiles as well as minimum, maximum, and mean
summary(minwageNJ$wageBefore)

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      4.25   4.25   4.50   4.61   4.87   5.75

summary(minwageNJ$wageAfter)

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      5.000   5.050   5.050   5.081   5.050   5.750

## interquartile range
IQR(minwageNJ$wageBefore)

## [1] 0.62

IQR(minwageNJ$wageAfter)

## [1] 0

## deciles (10 groups)
quantile(minwageNJ$wageBefore, probs = seq(from = 0, to = 1, by = 0.1))

##      0%   10%   20%   30%   40%   50%   60%   70%   80%   90%  100%
## 4.25 4.25 4.25 4.25 4.50 4.50 4.65 4.75 5.00 5.00 5.75

quantile(minwageNJ$wageAfter, probs = seq(from = 0, to = 1, by = 0.1))

##      0%   10%   20%   30%   40%   50%   60%   70%   80%   90%  100%
## 5.00 5.05 5.05 5.05 5.05 5.05 5.05 5.05 5.05 5.05 5.15 5.75
```

2.6.2: Standard Deviation

```
sqrt(mean((minwageNJ$fullPropAfter - minwageNJ$fullPropBefore)^2))

## [1] 0.3014669
```

```
mean(minwageNJ$fullPropAfter - minwageNJ$fullPropBefore)
```

```
## [1] 0.02387474
```

```
## standard deviation
```

```
sd(minwageNJ$fullPropBefore)
```

```
## [1] 0.2304592
```

```
sd(minwageNJ$fullPropAfter)
```

```
## [1] 0.2510016
```

```
## variance
```

```
var(minwageNJ$fullPropBefore)
```

```
## [1] 0.05311145
```

```
var(minwageNJ$fullPropAfter)
```

```
## [1] 0.0630018
```